# EXPERIMENT REPORT: RC CIRCUIT AS A FILTERING AND PHASE SHIFTING NETWORK

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## Abstract

This experiment investigates the behavior of RC circuits configured as low-pass and high-pass filters. The amplitude response and phase shift were measured over a range of input frequencies using a function generator and an oscilloscope. For both configurations, the experimental cut-off frequencies were found to be close to the theoretical value of $f_c \approx 305\,\text{Hz}$. Phase measurements confirmed the expected $0°$ to $\pm 90°$ variation across the frequency range. The results validate the theoretical transfer functions and demonstrate the effectiveness of RC networks as simple filtering and phase-shifting devices in signal processing applications.

# 1 Objective

- Study the transfer function and phase shift of a low pass RC filter network.
- Study the transfer function and phase shift of a high pass RC filter network.

# 2 Theory and working formula

A capacitor with a resistor in series can be used as a filter for high-frequency or low-frequency signals. Circuits that filter the high-frequency signals are known as high-pass filters whereas the circuits that filter the low-frequency signals are known as low-pass filters. Ideally, a low-pass filter will have a characteristic frequency $V_c$, known as the cut-off frequency, such that any signal with a higher frequency will be blocked. Similarly, there is a cut-off frequency $V_c$ for high-pass filters which will block any signal with a frequency less than it.

In practical filters, pass and stop bands are not clearly defined, $|\mathrm{H}(\mathrm{j}\omega)|$ varies continuously from its maximum towards zero. Therefore, the cut-off frequency is defined as the frequency at which $|\mathrm{H}(\mathrm{j}\omega)|$ is reduced to $1/\sqrt{2}$ or 0.7 of its maximum value which corresponds to signal power reduced by a factor of 2.

## 2.1 Low Pass Filters

A series RC circuit as shown in the circuit diagram in figure 1 also acts as a low-pass filter. For no load resistance (output is open circuit, $R \to \infty$):

$$V_o = \frac{V_i}{\sqrt{1+(\omega RC)^2}} = \frac{V_i}{\sqrt{1+(\frac{\omega}{\omega_c})^2}} \tag{1}$$

$$|\mathrm{H}(\mathrm{j}\omega)| = \frac{V_o}{V_i} = \frac{1}{\sqrt{1+(\frac{\omega}{\omega_c})^2}} \tag{2}$$

At the cut-off frequency ($\omega_c$), we have

$$|\mathrm{H}(\mathrm{j}\omega_\mathrm{c})| = \frac{1}{\sqrt{1+(\omega_c RC)^2}} = \frac{1}{\sqrt{2}} \tag{3}$$

The phase difference $\varphi$ is given by:

$$\varphi = -\arctan\frac{R}{\chi_c} = -\arctan(\omega RC) \tag{4}$$

The input impedence for the low-pass filter is given by:

$$Z_i = R + \frac{1}{j\omega C} \implies |Z_i| = \sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2} \tag{5}$$

The output impedance can be found by shorting the source and finding the equivalent impedance between the output terminals: $Z_0 = R||\frac{1}{j\omega C}$ where the source resistance is ignored.

## 2.2 High Pass Filters

For no load resistance (output open circuit), we have:

$$V_o = \frac{V_i}{\sqrt{1+(\frac{1}{\omega RC})^2}} = \frac{V_i}{\sqrt{1+(\frac{\omega_c}{\omega})^2}} \tag{6}$$

$$\mathrm{H}(\mathrm{j}\omega) = \frac{V_o}{V_i} = \frac{1}{\sqrt{1+(\frac{1}{\omega RC})^2}} = \frac{1}{\sqrt{1+(\frac{\omega_c}{\omega})^2}} \tag{7}$$

The phase difference $\varphi$ is given by equation 4.

The gain of this filter, $|\mathrm{H}(\mathrm{j}\omega)|$ is maximum when the denominator is smallest, i.e., $\omega \to \infty$, leading to $|\mathrm{H}(\mathrm{j}\omega)|_{\max} = 1$. Then, the cut-off frequency can be found as:

$$|\mathrm{H}(\mathrm{j}\omega_\mathrm{c})| = \frac{1}{\sqrt{1+(\frac{1}{\omega_c RC})^2}} = \frac{1}{\sqrt{2}} \tag{8}$$

Input and output impedences of this filter can be found similar to the procedure used for low-pass filters:

$$Z_i = \sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2} \quad \text{and} \quad Z_{i,min} = R \tag{9}$$

$$Z_o = R\,||\frac{1}{j\omega C} \quad \text{and} \quad Z_{o,max} = R \tag{10}$$

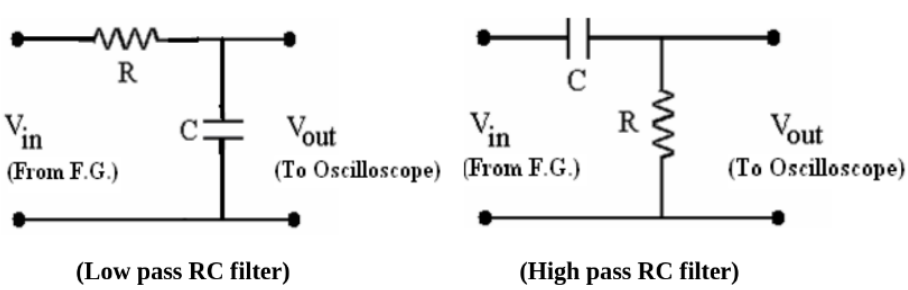



Figure 1: Circuit diagram for the low-pass and high-pass filter

## 3 Circuit Components/Instruments

(i) Resistor (5.6 kΩ), (ii) Capacitor (10 nF), (iii) Function generator, (iv) Oscilloscope, (v) Connecting wires, (vi) Breadboard

## 4 Observations

**R** = 5.538 kΩ, **C** = 94.1 nF

**(I) For Low Pass Filter:** $V_{in}$(pp) = 1.0 V, $\omega_c$= 1/RC = 1918 rad/s, $f_c$ = 305 Hz

Table 1: $|H(j\omega)|_{dB}$ and $\varphi$ data for the low-pass RC filter.

| Sl. No. | Frequency (Hz) | $V_o$(pp) (mV) | $\varphi$ (deg) | $\omega/\omega_c$ | $\lvert H(j\omega)\rvert$ | $\lvert H(j\omega)\rvert$ (theoretical) | $\varphi$ (theoretical) | $\lvert H(j\omega)\rvert_{dB}$ |
|---|---|---|---|---|---|---|---|---|
| 1 | 24.5 | 976 | -3 | 0.08 | 0.98 | 0.99 | -4.57 | -0.17 |
| 2 | 40.3 | 968 | -5 | 0.13 | 0.97 | 0.99 | -7.52 | -0.26 |
| 3 | 55.0 | 968 | -7 | 0.18 | 0.97 | 0.98 | -10.20 | -0.26 |
| 4 | 71.4 | 960 | -10 | 0.23 | 0.96 | 0.97 | -13.17 | -0.35 |
| 5 | 83.3 | 952 | -12 | 0.27 | 0.95 | 0.96 | -15.27 | -0.44 |
| 6 | 100.0 | 936 | -14 | 0.32 | 0.94 | 0.95 | -18.15 | -0.53 |
| 7 | 119.0 | 920 | -17 | 0.39 | 0.92 | 0.93 | -21.30 | -0.72 |
| 8 | 147.1 | 888 | -21 | 0.48 | 0.89 | 0.90 | -25.73 | -1.01 |
| 9 | 166.7 | 856 | -28 | 0.54 | 0.86 | 0.87 | -28.67 | -1.31 |
| 10 | 204.9 | 808 | -35 | 0.67 | 0.81 | 0.83 | -33.90 | -1.83 |
| 11 | 231.5 | 776 | -33 | 0.75 | 0.78 | 0.79 | -37.19 | -2.15 |
| 12 | 255.1 | 752 | -44 | 0.83 | 0.75 | 0.76 | -39.89 | -2.49 |
| 13 | 271.7 | 720 | -39 | 0.89 | 0.72 | 0.74 | -41.70 | -2.85 |
| 14 | 297.6 | 704 | -42 | 0.97 | 0.7 | 0.71 | -44.30 | -3.09 |
| 15 | 320.5 | 672 | -46 | 1.05 | 0.67 | 0.68 | -46.42 | -3.47 |
| 16 | 367.2 | 624 | -52 | 1.20 | 0.62 | 0.63 | -50.28 | -4.15 |
| 17 | 446.4 | 552 | -51 | 1.46 | 0.55 | 0.56 | -55.66 | -5.19 |
| 18 | 543.5 | 480 | -62 | 1.78 | 0.48 | 0.48 | -60.7 | -6.37 |
| 19 | 595.2 | 448 | -69 | 1.95 | 0.45 | 0.45 | -62.86 | -6.93 |

| Sl. No. | Frequency (Hz) | $V_o$(pp) (mV) | $\varphi$ (deg) | $\omega/\omega_c$ | $\lvert H(j\omega)\rvert$ | $\lvert H(j\omega)\rvert$ (theoretical) | $\varphi$ (theoretical) | $\lvert H(j\omega)\rvert_{dB}$ |
|---|---|---|---|---|---|---|---|---|
| 20 | 694.2 | 400 | -65 | 2.27 | 0.4 | 0.40 | -66.28 | -7.95 |
| 21 | 757.6 | 368 | -70 | 2.48 | 0.37 | 0.37 | -68.07 | -8.63 |
| 22 | 806.5 | 352 | -75 | 2.64 | 0.35 | 0.35 | -69.28 | -9.11 |
| 23 | 909.1 | 328 | -78 | 2.98 | 0.33 | 0.31 | -71.45 | -9.63 |
| 24 | 1000 | 296 | -79 | 3.27 | 0.3 | 0.29 | -73.04 | -10.45 |
| 25 | 1163 | 264 | -85 | 3.81 | 0.26 | 0.25 | -75.30 | -11.70 |
| 26 | 1289 | 232 | -81 | 4.22 | 0.23 | 0.23 | -76.68 | -12.76 |
| 27 | 1488 | 203 | -83 | 4.87 | 0.2 | 0.20 | -78.41 | -13.97 |
| 28 | 1603 | 192 | -87 | 5.25 | 0.19 | 0.18 | -79.22 | -14.42 |

**(II) For High Pass Filter:** $V_{in}$(pp) = 10.0 V, $\omega_c$= 1/RC = 1918 rad/s, $f_c$ = 305 Hz

Table 2: $|H(j\omega)|_{dB}$ and $\varphi$ data for the high-pass RC filter.

| Sl. No. | Frequency (Hz) | $V_o$(pp) (V) | $\varphi$ (deg) | $\omega_c/\omega$ | $\lvert H(j\omega)\rvert$ | $\lvert H(j\omega)\rvert$ (theoretical) | $\varphi$ (theoretical) | $\lvert H(j\omega)\rvert_{dB}$ |
|---|---|---|---|---|---|---|---|---|
| 1 | 2427 | 9.76 | -6 | 0.13 | 0.976 | 0.99 | -7.41 | -0.21 |
| 2 | 2212 | 9.76 | -9 | 0.14 | 0.976 | 0.99 | -7.97 | -0.21 |
| 3 | 2083 | 9.76 | -9 | 0.15 | 0.976 | 0.99 | -8.53 | -0.21 |
| 4 | 1724 | 9.76 | -9 | 0.18 | 0.976 | 0.98 | -10.2 | -0.21 |
| 5 | 1524 | 9.68 | -13 | 0.2 | 0.968 | 0.98 | -11.31 | -0.28 |
| 6 | 1412 | 9.6 | -14 | 0.22 | 0.96 | 0.98 | -12.41 | -0.35 |
| 7 | 1316 | 9.6 | -15 | 0.23 | 0.96 | 0.97 | -12.95 | -0.35 |
| 8 | 1059 | 9.52 | -19 | 0.29 | 0.952 | 0.96 | -16.17 | -0.43 |
| 9 | 932.8 | 9.44 | -21 | 0.33 | 0.944 | 0.95 | -18.26 | -0.5 |
| 10 | 776.4 | 9.2 | -24 | 0.39 | 0.92 | 0.93 | -21.31 | -0.72 |
| 11 | 710.2 | 9.12 | -26 | 0.43 | 0.912 | 0.92 | -23.27 | -0.8 |
| 12 | 529.7 | 8.56 | -31 | 0.58 | 0.856 | 0.87 | -30.11 | -1.35 |
| 13 | 378.8 | 7.68 | -38 | 0.81 | 0.768 | 0.78 | -39.01 | -2.29 |

| Sl. No. | Frequency (Hz) | $V_o$(pp) (V) | $\varphi$ (deg) | $\omega_c/\omega$ | $\lvert H(j\omega)\rvert$ | $\lvert H(j\omega)\rvert$ (theoretical) | $\varphi$ (theoretical) | $\lvert H(j\omega)\rvert_{dB}$ |
|---|---|---|---|---|---|---|---|---|
| 14 | 250.0 | 6.4 | -54 | 1.22 | 0.64 | 0.63 | -50.66 | -3.88 |
| 15 | 138.9 | 4.24 | -65 | 2.2 | 0.424 | 0.41 | -65.56 | -7.45 |
| 16 | 123.8 | 3.84 | -65 | 2.46 | 0.384 | 0.38 | -67.88 | -8.31 |
| 17 | 104.2 | 3.36 | -67 | 2.93 | 0.336 | 0.32 | -71.16 | -9.47 |
| 18 | 87.7 | 2.88 | -75 | 3.48 | 0.288 | 0.28 | -73.97 | -10.81 |
| 19 | 68.5 | 2.24 | -77 | 4.45 | 0.224 | 0.22 | -77.33 | -13.0 |
| 20 | 47.2 | 1.76 | -83 | 6.46 | 0.176 | 0.15 | -81.2 | -15.09 |
| 21 | 43.9 | 1.52 | -82 | 6.95 | 0.152 | 0.14 | -81.81 | -16.36 |
| 22 | 28.4 | 0.91 | -80 | 10.74 | 0.091 | 0.09 | -84.68 | -20.82 |
| 23 | 24.5 | 0.8 | -81 | 12.45 | 0.08 | 0.08 | -85.41 | -21.94 |
| 24 | 21.7 | 0.71 | -85 | 14.06 | 0.071 | 0.07 | -85.93 | -22.97 |
| 25 | 19.1 | 0.63 | -83 | 15.97 | 0.063 | 0.06 | -86.42 | -24.01 |

# 5 Results

By fitting for the low pass filter dB data with respect to angular frequency using equation 1, the critical frequency is obtained as:

$$\boxed{\omega_c = (1909.99 \pm 156.22)\text{rad/s}}$$

The percentage error in the value of the critical frequency from the theoretical value for the low-pass filter:

$$\begin{aligned}\%\,\text{error} &= \left|\frac{\text{experimental value} - \text{theoretical value}}{\text{theoretical value}}\right| \times 100\% \\ &= 0.33\%\end{aligned}$$

For the high frequency filter, we fit using equation 6 to get:

$$\boxed{\omega_c = (1859.35 \pm 304.15)\text{rad/s}}$$

The percentage error in the value of the critical frequency from the theoretical value for the high-pass filter:

$$\begin{aligned}\%\,\text{error} &= \left|\frac{\text{experimental value} - \text{theoretical value}}{\text{theoretical value}}\right| \times 100\% \\ &= 2.97\%\end{aligned}$$

The graph plots for the $|\mathrm{H(j\omega)}|_{\mathrm{dB}}$ and $\varphi$ for the low-pass and the high-pass circuits are given in figures 2 and 3 with the fitting curve (for $|\mathrm{H(j\omega)}|_{\mathrm{dB}}$ plot) and the curve with theoretical values (for $\varphi$ plot).

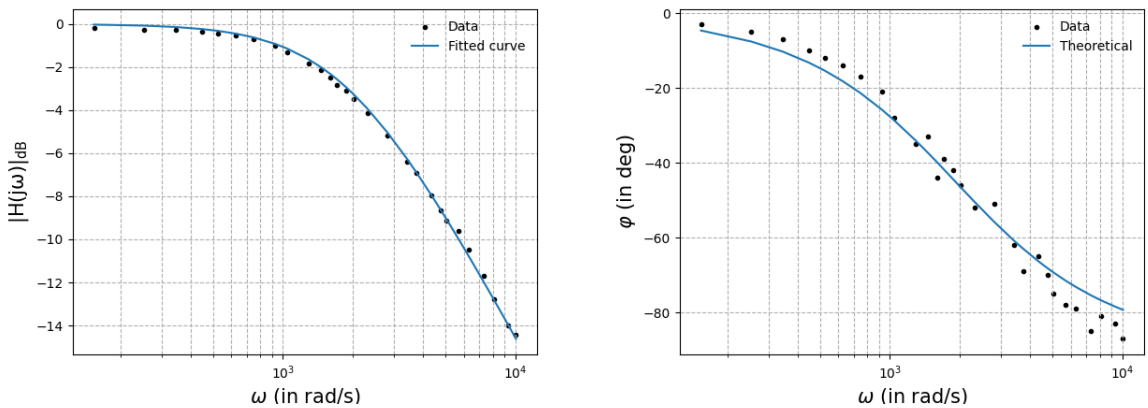
# 6 Discussion

From the experimental data and the fitted curves, the values of the cut-off frequency $\omega_c$ for both the low-pass and high-pass filters are found to be close to the theoretical value $\omega_c = 1/RC \approx 1918\,\text{rad/s}$, with small deviations within the experimental uncertainty. This agreement validates the theoretical prediction that the $-3$ dB point corresponds to the cut-off frequency.

For the low-pass filter, the gain was nearly unity at low frequencies and gradually decreased beyond $f_c \approx 305\,\text{Hz}$. The measured phase shift also showed the expected trend: close to $0°$ at low frequencies, approaching $-90°$ at high frequencies, with approximately $-45°$ at the cut-off frequency.

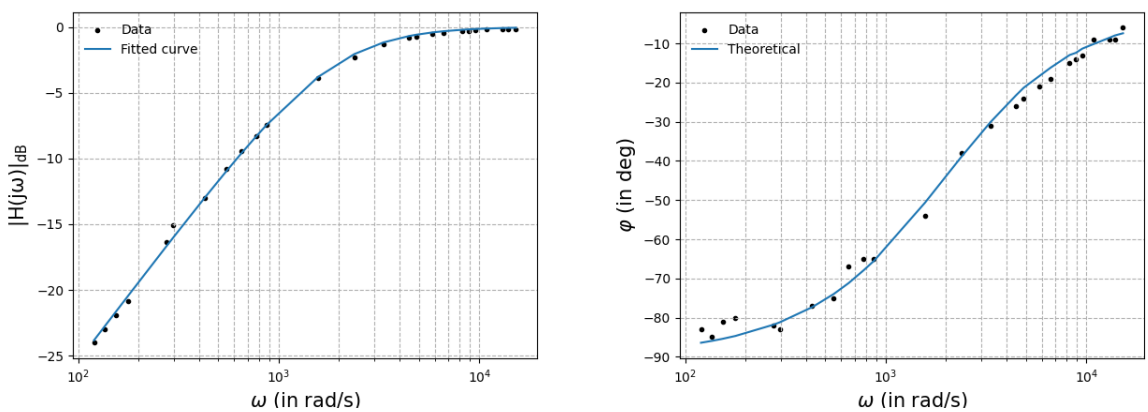
For the high-pass filter, the gain was suppressed at low frequencies and gradually increased toward unity as frequency increased, again consistent with theory. The phase shift varied from about $-90°$ at very low frequencies to $0°$ at high frequencies, with a $-45°$ shift around the cut-off frequency. Experimental data showed close agreement with this expected trend, though a slightly higher scatter in phase measurements was observed.

Overall, the experiment successfully demonstrated the filtering and phase-shifting properties of RC networks. The experimental uncertainties are consistent with instrumental limitations, and no significant systematic deviation from theory was observed.




(a) $|\mathrm{H}(\mathrm{j}\omega)|_{\mathrm{dB}}$ plot (b) Phase difference $\varphi$ plot

Figure 2: Bode plots for the low-pass RC circuit for $|\mathrm{H}(\mathrm{j}\omega)|_{\mathrm{dB}}$ and phase difference $\varphi$




(a) $|\mathrm{H}(\mathrm{j}\omega)|_{\mathrm{dB}}$ plot (b) Phase difference $\varphi$ plot

Figure 3: Bode plots for the high-pass RC circuit for $|\mathrm{H}(\mathrm{j}\omega)|_{\mathrm{dB}}$ and phase difference $\varphi$

# 7 Conclusion

The experiment verified the theoretical behavior of RC circuits as both low-pass and high-pass filters. The experimentally determined cut-off frequencies for the two circuits were in close agreement with the theoretical value of $f_c \approx 305\,\mathrm{Hz}$ ($\omega_c \approx 1918\,\mathrm{rad/s}$), within the error bars of the experimental value. For the low-pass filter, the experimental value of critical frequency is within 0.33% error from the theoretical value whereas for the high-pass filter it is within 2.97% error from the theoretical value of the critical frequency.

Thus, the experiment confirms that simple RC networks can be effectively used as frequency-selective filters and phase-shifting elements in signal processing.

# 8 Precautions

1. Ensure that the ground of the oscilloscope, function generator and the probes are in the same location (that is, they are shorted).

2. Avoid non-insulated parts of the wires/probe to come in contact with each other throughout the experiment.

3. Ensure that the voltage (pp) generated by the function generator is constant throughout the experiment.